



UE24MA241B LINEAR ALGEBRA AND ITS APPLICATIONS

Dr. Gayathri T

Department of Computer Science and
Engineering

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION

slides by Renna Sultana, Department of Science and Humanities



LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:



Course Content: Gaussian Elimination

❖ **Rank Of A Matrix:** A Square matrix A of order n is said to have rank r if

- At least one minor of order r does not vanish ($\neq 0$).
- Every minor of order (r+1) vanishes ($= 0$).

Rank of matrix A is denoted by r i.e. rank(A)=r.

❖ If $A = \begin{bmatrix} a_{ij} \end{bmatrix}_{m \times n}$ is a rectangular matrix, then **Rank of the Matrix** is defined as the number of non-zero rows in the Echelon form of A. It is also defined as the maximum number of Linearly Independent Rows or Columns of the Matrix A.

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:



$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 0 \end{pmatrix}; \begin{pmatrix} 2 & 2 & -3 \\ -2 & -2 & 3 \\ 4 & 4 & 6 \end{pmatrix}; \begin{pmatrix} 1 & 5 \\ -1 & 3 \\ 2 & 3 \end{pmatrix}; \begin{pmatrix} 1 & 5 \\ -1 & 3 \\ 2 & 10 \end{pmatrix}; \begin{pmatrix} 2 & -1 & 3 & 5 \\ 3 & 2 & 1 & 2 \end{pmatrix}$$

↓

2

↓

1

↓

2

↓

2

↓

2

Ex: Find the conditions on a and b so that the Matrix has rank 1, 2, 3.

$$\begin{pmatrix} a & 1 & 2 \\ 0 & 2 & b \\ 1 & 3 & 6 \end{pmatrix} \text{ has}$$

$$\begin{pmatrix} a & 1 & 2 \\ 0 & 2 & b \\ 1 & 3 & 6 \end{pmatrix} \xrightarrow{R_3 - \left(\frac{1}{a}\right)R_1} \begin{pmatrix} a & 1 & 2 \\ 0 & 2 & b \\ 0 & 3 - (1/a) & 6 - (2/a) \end{pmatrix}$$

- (i) For no values of a and b this matrix will have rank 1.
- (ii) If $a=1/3$ and $b=4$, rank of the matrix is 2.
- (iii) If $a \neq 1/3$ and $b \neq 4$, rank of the matrix is 3.

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:



❖ Relation between Rank, Consistency and Solution:

❖ If $\text{rank}(A)=r$, then the following hold good:

- (i) If $\text{rank}(A)=\text{rank}[A:b]=r$, system $Ax=b$ is **consistent** and has **a solution**.
- (ii) If $\text{rank}(A)=\text{rank}[A:b]=r=n$, system $Ax=b$ is **consistent** and has **a unique solution**.
- (iii) If $\text{rank}(A)=\text{rank}[A:b]=r<n$, system $Ax=b$ is **consistent** and has **infinite number of solutions**.
- (iv) If $\text{rank}(A) \neq \text{rank}[A:b]$, system $Ax=b$ is **inconsistent** and has **no solution**.

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:



Gaussian Elimination

Gaussian Elimination is used to check for Consistency and solve a System of linear equations.

For a given system of equations $Ax=b$ apply Elementary row transformations to the Augmented Matrix $[A:b]$ and reduce it to $[U:c]$ where U is an **Upper Triangular Matrix** so that we get an equivalent system $Ux=c$ which can be solved by **Backward Substitution**.

Here A and U are Equivalent Matrices and hence solution of $Ax=b$ is same as $Ux=c$.

❖ **The following steps are to be followed while performing Elementary Row Transformations in Gaussian Elimination:**

- **No exchange** of rows.
- First row should be **retained** as it is (**not altered**).
- The first non-zero element in every non-zero row is called **Pivot**.
- The original system $Ax=b$ and new system obtained $Ux=c$ have the **same solution**.

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:



Gaussian Elimination is illustrated below for a system of 3 equations with 3 variables.

Consider a system of 3 equations in 3 variables $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$[A : b] = \begin{pmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ a_{21} & a_{22} & a_{23} : b_2 \\ a_{31} & a_{32} & a_{33} : b_3 \end{pmatrix} \xrightarrow{\substack{R_2 - \left(\frac{a_{21}}{a_{11}}\right)R_1 \\ R_3 - \left(\frac{a_{31}}{a_{11}}\right)R_1}} \begin{pmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ 0 & d_{22} & d_{23} : c_2 \\ 0 & d_{32} & d_{33} : c_3 \end{pmatrix}$$

$$\xrightarrow{R_3 - \left(\frac{d_{32}}{d_{22}}\right)R_2} \begin{pmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ 0 & d_{22} & d_{23} : c_2 \\ 0 & 0 & e_{33} : c_4 \end{pmatrix} = [U : c]$$

$$\Rightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ d_{22}x_2 + d_{23}x_3 = c_2 \\ e_{33}x_3 = c_4 \end{cases}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:

Check for consistency and solve the following system of equations if consistent:

$$\begin{aligned} \text{(i)} \quad & x_1 + x_2 - 2x_3 + 4x_4 = 5 \\ & 2x_1 + 2x_2 - 3x_3 + x_4 = 3 \\ & 3x_1 + 3x_2 - 4x_3 - 2x_4 = 1 \end{aligned} \quad [A:b] = \begin{pmatrix} 1 & 1 & -2 & 4 & : & 5 \\ 2 & 2 & -3 & 1 & : & 3 \\ 3 & 3 & -4 & -2 & : & 1 \end{pmatrix}$$

$$\xrightarrow[R_3-3R_1]{R_2-2R_1} \begin{bmatrix} 1 & 1 & -2 & 4 & : & 5 \\ 0 & 0 & 1 & -7 & : & -7 \\ 0 & 0 & 2 & -14 & : & -14 \end{bmatrix} \xrightarrow[R_3]{R_2 \nrightarrow R_1} \begin{bmatrix} 1 & 1 & -2 & 4 & : & 5 \\ 0 & 0 & 1 & -7 & : & -7 \\ 0 & 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + x_2 - 2x_3 + 4x_4 = 5 \\ x_3 - 7x_4 = -7 \end{cases} \quad \begin{array}{l} r(A)=2 = r[A:b] < n(=4) \\ \text{System is consistent and has infinitely many solutions.} \end{array}$$

Solution is $(x_1, x_2, x_3, x_4) = (10k_1 - k_2 - 9, k_2, 7k_1 - 7, k_1)$

Depending upon values of k_1 and k_2 we get infinity of solutions.



THANK YOU

Dr. Gayathri T

Department of Computer Science and Engineering

gayathrit@pes.edu